DO YOU KNOW?



Universal motors are known to sometimes over-speed, perhaps dangerously and destructively, when run with no load. This article deals with different methods of universal electric motor speed control.

A motor which can be operated either on DC (direct current) or on single phase AC (alternating current) is called a universal motor. On both AC and DC, it gives approximately the same speed and output. Universal motors often run at dangerously high speed when operated at no load. This means that motor speed will be low at full load, but the speed of the motor will start increasing as the load on the motor decreases. Finally at no load, the motor will run at very high speed (perhaps about 20,000 rpm in some cases) and cannot made to run at less than about 2,000 rpm. The speed of universal motor varies just like that of a DC series motor. Being a series wound motor, it has a high starting torque and variable speed characteristics, and if the motor is started at no load, due to its high starting torque it will attain high speed quickly. Thus it is not advisable to start a universal motor at no load.

7.6 APPLICATIONS OF DC GENERATORS

1. DC Series Generators

i) They are used for series arc lighting
ii) They are used as boosters, for the purpose of compensating the drop in voltage in the lines on loading.
iii) Used for regenerative braking of DC locomotives.

2. DC Shunt Generators are Used

i) In shunt generator, terminal voltage is more or less constant. So these are used for supplying loads needing constant voltage.
ii) In Electroplating
iii) In Battery charging purpose.
iv) As Exciters for AC generators.

3. DC Compound Generators

i) By means of compound generator it is possible to give constant voltage at the line end by proper compounding.
ii) Differentially, compound generator may be used for welding purpose.
iii) They are used to supply power to railway circuits, elevator motors etc.

4. Separately Excited Generators

These generators are used for:

i) Supplying DC motors whose speed is to be varied widely.
ii) Where a wide range of DC voltage is required for testing purpose.

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